A Replicated Log, Set in Two Columns

A two-page fixture for the reading order of columns and tables

1 Introduction

A replicated log keeps the same sequence of records on several machines, so that losing one machine loses no record. Each record is appended by a leader and copied to the followers before it counts as written.

This fixture prints that idea in two columns, the way most papers are printed. Both columns share every baseline, so a reader that sorts the page by height alone reads a line of the left column, then a line of the right one, and gives itself away.

2 Roles

Table 1 lists the roles in a group of five replicas. Only the leader accepts appends; the followers copy and acknowledge them, and a learner copies without a vote.

Table 1: Replica roles in a group of five.

Role	Count	Votes
Leader	1	yes
Follower	3	yes
Learner	1	no

A write is durable once a majority of the voting replicas holds it. With four voters, three acknowledgements are enough, so one slow follower never delays a commit, and the leader can answer the client as soon as the third one arrives.

Followers learn of new records in batches. The leader sends each follower the records after the last one it acknowledged, and a follower that has fallen too far behind is sent a snapshot instead of the records it missed.

3 Recovery

When the leader stops answering, each follower waits out a random timeout and then asks the others for their votes. The first to win a majority becomes the new leader and goes on from the last record a majority holds.

A record that only the old leader held is dropped. That is safe, because no client was ever told that the record had been written.

Table 2: Write latency and throughput by group size.

Replicas	p50 write (ms)	p99 write (ms)	Writes/s
3	1.8	4.9	61,000
5	2.3	6.1	54,000
7	2.9	7.7	47,000

4 Evaluation

Table 2 shows that each pair of replicas added to a group costs about half a millisecond at the median. The extra replicas buy tolerance of more failures, not speed.

Throughput falls more slowly than latency rises, because the leader batches the appends that arrive while it waits for acknowledgements, and a larger group waits a little longer and so sends larger batches.

5 Lessons

Keep the voting group small and add learners for reads. Measure the slowest follower that a majority must wait for, not the average one, since that follower sets the latency of every write.